Name: ____________________________ Class: ________________

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CATHOLIC JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1
JC2 Preliminary Exam

MATHEMATICS 8864/01

Paper 1 **27 August 2013**

3 hours

Additional Materials: List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, arrange your answers in NUMERICAL ORDER.
Place this cover sheet in front and fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

Question	Marks
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
Total / 95	
Calculator Model No.	

Section A: Pure Mathematics [35 Marks]

1. Find, algebraically, the set of values of k for which $kx^2 + kx - 2x + k < 0$ for all real values of x. [4]

2. Find $\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{5}{2}\ln x + \sqrt{x^5+4}\right]$ and hence evaluate exactly $\int_1^2 \left(\frac{1}{x} + \frac{x^4}{\sqrt{x^5+4}}\right)\mathrm{d}x$. [4]

3. **(i)** On a single diagram, sketch the graphs of $y = \frac{1}{x-10}$ and $y = \ln(2x+1)$, stating clearly the coordinates of all points of intersection with the axes, and equations of asymptotes. [3]

(ii) State the coordinates of the points where curves $y = \frac{1}{x-10}$ and $y = \ln(2x+1)$ intersect. [1]

(iii) Hence or otherwise, find the set of values of x for which $\frac{1}{2(x-10)} < \ln\sqrt{2x+1}$. [2]

Hence solve the inequality $\frac{1}{2(x-9)} < \ln\sqrt{2x+3}$. [2]

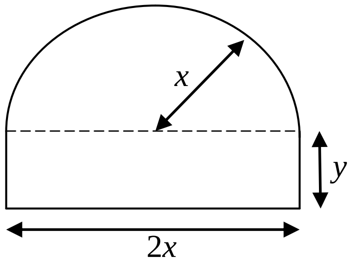
4. A curve has gradient given by $4x - \frac{16}{x^2}$.

(i) If the curve passes through $Q(2, 16)$, show that the equation of the curve is $y = 2x^2 + \frac{16}{x}$. [3]

(ii) Find the equation of the normal to the curve at the point $Q(2, 16)$. [3]

(iii) Find the area bounded by the curve, the normal at the point Q, the line $x = 1.5$ and the x-axis, giving your answer correct to one decimal place. [3]

5. A school has a big window in the school hall. The shape of the window consists of a semi-circle of radius x metres on top of a rectangle, $2x$ metres by y metres, as shown in the diagram.



(i) If the area of the window is fixed at 10 m^2, show that its perimeter, P, is $\frac{10}{x} + \frac{\pi x}{2} + 2x$. [3]

The school wants to install a special netting over the window to keep out dust particles brought by haze. An expensive glue is applied along the perimeter, P, of the window to attach the netting.

(ii) Find the exact value of x for which P is a minimum. [3]

A machine is used to apply the glue. The glue is poured into and applied by the machine at a constant rate of 40 cm^3 per minute. The amount of glue applied by the machine is given by $V = \frac{1}{27}\pi w^3$ cm^3 where w is the perimeter of the window applied with glue by the machine.

(iii) Find the rate of increase of w at the instant when $w = 20$ cm. [4]

Section B: Statistics [60 Marks]

6. A hospital has the following number of patients in four different wards during the week when the Pollutant Standards Index reading was between 300 to 400.

	Cancer Ward	Geriatric Ward	Pediatrics Ward	Maternity Ward
Number of patients	390	360	340	410

In order to assess the impact of air pollution on its patients, the hospital decides to survey a random sample of 100 patients.

(i) Explain what is meant in this context by the term 'a random sample'. [1]

(ii) Describe clearly how a sample of 100 patients could be chosen using stratified sampling. [2]

(iii) Give one advantage and one disadvantage of using stratified sampling over quota sampling in this context. [2]

7. Events A and B are such that $\mathrm{P}(A)=\frac{1}{3}$, $\mathrm{P}(B)=\frac{1}{4}$ and $\mathrm{P}(A'\cap B)=\frac{7}{60}$.

Find **(i)** $\mathrm{P}(A\cap B)$, [2]

(ii) $\mathrm{P}(A'\,|\,B')$. [3]

State, with a reason in each case, whether A and B are independent, and whether A and B are mutually exclusive. [2]

8. Trainee teachers must pass an examination before they can graduate from the National Institute of Education (NIE). A trainee teacher who fails the examination at the first attempt is allowed a maximum of two more attempts. The probability that a randomly chosen trainee teacher passes the examination at the first attempt is $\frac{3}{5}$. For each attempt after the first, the probability of passing the examination is $\frac{4}{5}$.

(i) Draw a tree diagram to represent the possible outcomes. [2]

(ii) Find the probability that a randomly chosen trainee teacher graduates from NIE. [3]

(iii) Three trainee teachers are chosen at random. Find the probability that one of them passes at the first attempt and the other two pass at the third attempt. [3]

9. The probability that a flight is delayed at an airport is 15%. Assume that flight delays occur independently.

(i) Find the probability that out of 10 flights,

(a) exactly 2 are delayed, [1]

(b) more than 4 are delayed. [2]

(ii) Find the maximum number of flights such that the probability of at least one flight being delayed is less than 0.9. [3]

(iii) 50 random flights are recorded. Use a suitable approximation to estimate the probability that fewer than 42 flights are not delayed. [3]

10. To study the recent haze situation, a student from the Environmental Club recorded the number of hotspots in Sumatra, x, and the corresponding Pollutant Standards Index (PSI), y, over a period of eight days using data from the National Environment Agency. The results are summarised in the table below.

Date	18 Jun	19 Jun	23 Jun	24 Jun	25 Jun	26 Jun	27 Jun	28 Jun
x	187	173	227	437	174	59	42	15
y	128	107	98	89	77	62	59	60

(i) Draw a sketch of the scatter diagram for the data, as shown on your calculator. [1]

(ii) Find the product moment correlation coefficient. [1]

(iii) Identify a data pair which should be regarded as suspect. Remove the suspect data pair, and calculate the product moment correlation coefficient for the revised data. Comment on its value with your answer to **(ii)** in the context of the data. [3]

(iv) Find the equation of the regression lines of y on x, and x on y, for the revised data. [2]

(v) Using a suitable regression line obtained in **(iv)**, estimate the number of hotspots when the PSI is 300, giving a reason for your choice of regression line. Comment on the reliability of this estimate. [3]

11. The average waiting time at a polyclinic under usual conditions is 28 minutes. During the haze season, the polyclinic receives more patients seeking medical treatment for respiratory illness. As a result, patients complain that the waiting time is longer. To investigate this complaint, the waiting times, t minutes, of 100 randomly selected patients are recorded and the results are summarised by $\sum(t-30)=-117$ and $\sum(t-30)^2=1532$.

(i) Find unbiased estimates of the population mean and variance. [2]

(ii) What do you understand by the term 'unbiased estimate'? [1]

(iii) Test at the 3% significance level whether the complaint is valid.
Hence, state the lowest level of significance for which the complaint is valid [4]

(iv) Another random sample of n patients gives a mean waiting time of 30 minutes. Given that the population standard deviation is 14 minutes, find the largest possible value of n such that the complaint is not valid at the 3% level of significance. [4]

12. A local entrepreneur capitalises on a city's haze situation by importing two models of air purifiers, A and B. The weights, in kilograms, of the two models of air purifiers have independent normal distributions. The means and standard deviations of these distributions, and the shipping cost, in \$ per kilogram, are shown in the following table.

	Mean (kg)	Standard deviation (kg)	Shipping cost ($ per kg)
Model A	16	1.2	5
Model B	35	2.5	7

Stating clearly the mean and variance of all distributions that you use, find the probability that

(i) the total weight of 6 randomly chosen air purifiers of model A is less than 100 kg. [2]

(ii) the total shipping cost of 6 randomly chosen air purifiers of model A exceeds twice the shipping cost of one randomly chosen air purifier of model B. [4]

A random sample of 50 air purifiers of model A and a random sample of 20 air purifiers of model B are taken.

(iii) Find the probability that the difference in mean weights between the two samples is less than 20 kg. [4]

— END OF PAPER —